

Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 2

Convergence of Computational Algorithms

Sindri Magnússon

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Overview

- 1 Introduction
- 2 Contractive Algorithms
- 3 Linear Algorithms
- 4 Stochastic Algorithms

Computational Algorithms

Many algorithms are on the form

$$x_{k+1} = T(x_k)$$

or

$$x_{k+1} = T(x_k) + n_k$$

where n_k is some noise or error.

Computational Algorithms: Examples

For optimization problems

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x)$$

- Gradient descent

$$x_{k+1} = T(x_k) := x_k - \gamma \nabla f(x_k)$$

- Newtons Method

$$x_{k+1} = T(x_k) := x_k - \gamma H(x_k)^{-1} \nabla f(x_k)$$

Computational Algorithms: Examples

For optimization problems

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x)$$

- Polyak's Heavy ball method (GD with momentum)

$$x_{k+1} = x_k - \gamma \nabla f(x_k) + \beta(x_k - x_{k-1})$$

Computational Algorithms: Examples

For optimization problems

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x)$$

- Polyak's Heavy ball method (GD with momentum)

$$x_{k+1} = x_k - \gamma \nabla f(x_k) + \beta(x_k - x_{k-1})$$

By setting $z_k = (x_k, x_{k-1})$ we get

$$z_{k+1} = T(z_k) := \begin{pmatrix} x_k - \gamma \nabla f(x_k) + \beta(x_k - x_{k-1}) \\ x_k \end{pmatrix}$$

Computational Algorithms: Examples

For optimization problems

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x)$$

- Stochastic Gradient

$$x_{k+1} = x_k - \gamma \nabla f(x_k) + n_k$$

where n_k are i.i.d. unbiased noise with finite variance, i.e.,
 $\mathbb{E}n_k = 0$ and $\mathbb{E}||n_k||^2 < \infty$

Computational Algorithms: Examples

For optimization problems

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x) = \sum_{n=1}^N f_n(x)$$

- Stochastic Gradient

$$x_{k+1} = x_k - \gamma \nabla f_{n_k}(x_k)$$

where n_k is selected uniformly from $\{1, 2, \dots, N\}$

Computational Algorithms: Examples

For optimization problems

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x) = \sum_{n=1}^N f_n(x)$$

- Stochastic Gradient

$$x_{k+1} = x_k - \gamma \nabla f_{n_k}(x_k)$$

where n_k is selected uniformly from $\{1, 2, \dots, N\}$

- Minibatch SGD

$$x_{k+1} = x_k - \gamma \frac{1}{|J_k|} \sum_{n \in J_k} \nabla f_n(x_k)$$

where J_k is uniformly random subset of $\{1, 2, \dots, N\}$.

Computational Algorithms: Examples

Many other example

- Dynamic programming
- TD-learning (e.g., Q-learning)
- Game dynamics and economics
- Control dynamics
- Many distributed Algorithms
- Communications and networking

Computational Algorithms

General Rule

Algorithm

$$x_{t+1} = T(x_t)$$

can only converge to one of its fixed points, i.e., to x^* such that $x^* = T(x^*)$. **But the system is not guaranteed to converge.**

- Fixed points
- Equilibrium points

Computational Algorithms

General Rule

$$x_{t+1} = T(x_t)$$

can only converge to one of its fixed points, i.e., to x^* such that $x^* = f(x^*)$. **But the system is not guaranteed to converge.**

- Do fixed points always exist?
- If dynamics have fixed point do we converge to them?
- If they exist are they unique?
- What happens when we have noise in our algorithms/systems?

Computational Algorithms

Question 1

Do fixed point always exist?

Computational Algorithms

Question 1

Do fixed point always exist?

No, generally not. Considered $T(x) = x + 1$

- $x = T(x)$ would imply $x = x + 1$ or $0 = 1$.

Computational Algorithms

Question 1

Do fixed point always exist?

Brouwer fixed-point theorem

Consider a continuous function $T : \mathcal{X} \rightarrow \mathcal{X}$ where $\mathcal{X}$ is a compact and convex subset of $\mathbb{R}^d$. Then T has a fixed point, i.e., there exists $x \in \mathcal{X}$ such that $x = T(x)$.

There exists hundreds of similar Fixed-point theorems

Computational Algorithms

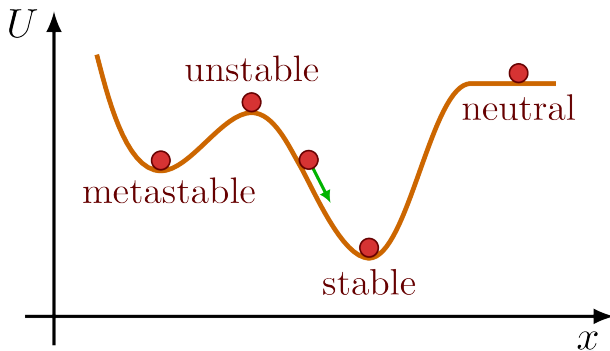
Question 2

If dynamics have a fixed point, do we converge to them?

Computational Algorithms

Question 2

If dynamics have a fixed point, do we converge to them?



Overview

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Contractive Algorithms

Definition: Lipschitz Continuity

Consider a function $T : \mathcal{X} \rightarrow \mathcal{X}$ and let $\|\cdot\|$ be some norm on $\mathcal{X}$. We say that T is L -Lipschitz continuous if

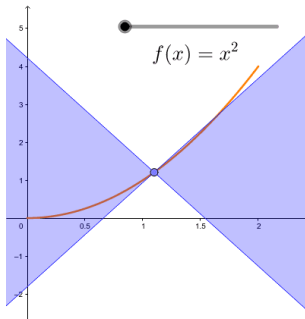
$$\|T(x) - T(y)\| \leq L\|x - y\| \text{ for all } x, y \in \mathcal{X}$$

Contractive Algorithms

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Contractive Algorithms

Definition: Contractive mapping

An α -Lipschitz continuous function $T : \mathcal{X} \rightarrow \mathcal{X}$ is called contraction mapping if $\alpha \in [0, 1)$.

Contractive Algorithms

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An α -Lipschitz continuous function $T : \mathcal{X} \rightarrow \mathcal{X}$ is called contraction mapping if $\alpha \in [0, 1)$.

Brouwer fixed-point theorem

If $T : \mathcal{X} \rightarrow \mathcal{X}$ is an α -contraction then T has a unique fixed point $x^* \in \mathcal{X}$ and $\lim_{t \rightarrow \infty} T^t(x_0) = x^*$ for all $x_0 \in \mathcal{X}$.

Contractive Algorithms

Definition: pseudo contractive mapping

We call a function $T : \mathcal{X} \rightarrow \mathcal{X}$ an α -pseudo contraction if $\alpha \in [0, 1)$ and there exists $x^* \in \mathcal{X}$ such that

Contractive Algorithms

Definition: pseudo contractive mapping

We call a function $T : \mathcal{X} \rightarrow \mathcal{X}$ an α -pseudo contraction if $\alpha \in [0, 1)$ and there exists $x^* \in \mathcal{X}$ such that

Theorem

α -contraction mapping is also α -pseudo contractive mapping.

Contractive Algorithms

- For an α -contractive algorithm $x_{k+1} = T(x_k)$ we have

$$\begin{aligned} \|x_{k+1} - x^*\| &= \|T(x_k) - T(x^*)\| \leq \alpha \|x_k - x^*\| \\ &\leq \alpha^{k+1} \|x_0 - x^*\| \end{aligned}$$

Contractive Algorithms

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- Converges at exponential (or α -linear) rate.

Contractive Algorithms

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- Converges at exponential (or α -linear) rate.
- We find an ϵ -accurate solution, $\|x_0 - x^*\| \leq \epsilon$, after

$$k \approx \frac{1}{1 - \alpha} \log \left(\frac{\|x_0 - x^*\|}{\epsilon} \right) \text{ iterations.}$$

Contractive Algorithms: Example

Consider optimization problems

$$\underset{x \in \mathbb{R}^d}{\text{minimize}} \quad f(x)$$

Assumptions: L -smooth and μ -convex

Suppose that f is L -smooth:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \text{for all } x, y \in \mathbb{R}^d$$

and μ -convex

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|x - y\|^2 \quad \text{for all } x, y \in \mathbb{R}^d$$

Contractive Algorithms: Example

Take $\gamma \in (0, 2/L)$ and consider

$$T(x) = x - \gamma \nabla f(x).$$

Then $T(x)$ is α pseudo-contraction with

$$\alpha = \max\{|1 - \gamma\mu|, |1 - \gamma L|\}.$$

Contractive Algorithms: Example

In particular, if we set $\gamma = 1/L$ follow the algorithm

$$x_{k+1} = x_k - \gamma \nabla f(x_k)$$

then we get

$$\|x_k - x^*\| \leq \left(1 - \frac{1}{\kappa}\right)^k \|x_0 - x^*\|$$

where the ratio $\kappa = L/\mu$ is called the **condition number** of the problem.¹

¹For quadratic problems, κ will be the ratio between the largest and smallest eigenvalues.

Contractive Algorithms: Example

Iteration Complexity

$$\|x_k - x^*\| \leq \left(1 - \frac{1}{\kappa}\right)^k \|x_0 - x^*\|$$

means that we find an ϵ -accurate solution, $\|x_k - x^*\| \leq \epsilon$, after

$$k \approx \kappa \log \left(\frac{\|x_0 - x^*\|}{\epsilon} \right)$$

Contractive Algorithms: Example

Iteration Complexity

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means that we find an ϵ -accurate solution, $\|x_k - x^*\| \leq \epsilon$, after

$$k \approx \kappa \log \left(\frac{\|x_0 - x^*\|}{\epsilon} \right)$$

With Nesterov's accelerated gradient we find ϵ -solution in

$$k \approx \sqrt{\kappa} \log \left(\frac{\|x_0 - x^*\|}{\epsilon} \right)$$

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Linear Algorithms

Consider an algorithm like

$$x_{k+1} = Ax_k + b$$

- Do fixed points exist?
- If they exist are they unique?
- If dynamics have fixed point do we converge to them?

Linear Algorithms

Consider an algorithm like

$$x_{k+1} = Ax_k + b$$

Do fixed points exist?

Linear Algorithms

Consider an algorithm like

$$x_{k+1} = Ax_k + b$$

Do fixed points exist? Only if the following linear system has a solution

$$x = Ax + b \iff (I - A)x = b$$

each solution is a fixed point of the algorithm.

Linear Algorithms

Consider an algorithm like

$$x_{k+1} = Ax_k + b$$

When does the system have a unique fixed point?

Linear Algorithms

Consider an algorithm like

$$x_{k+1} = Ax_k + b$$

When does the system have a unique fixed point? When the system $(I - A)x = b$ has a unique solution:

- If the system is overdetermined we will have either unique solution or
- If $I - A$ we have the unique solution $x^* = (I - A)^{-1}b$.

Linear Algorithms

Consider an algorithm like

$$x_{k+1} = Ax_k + b$$

Does the algorithm converge to some of its fixed points?

Linear Algorithms

Consider an algorithm like

$$x_{k+1} = Ax_k + b$$

Does the algorithm converge to some of its fixed points? We have that

$$\begin{aligned} x_{k+1} &= Ax_k + b = A(Ax_{k-1} + b) + b = A^2x_{k-1} + Ab + b \\ &= A^{k+1}x_0 + \left(\sum_{i=0}^k A^i \right) b \end{aligned}$$

Linear Algorithms

Consider an algorithm like

$$x_{k+1} = Ax_k + b$$

Does the algorithm converge to some of its fixed points? We have that

$$\begin{aligned} x_{k+1} &= Ax_k + b = A(Ax_{k-1} + b) + b = A^2x_{k-1} + Ab + b \\ &= A^{k+1}x_0 + \left(\sum_{i=0}^k A^i \right) b \end{aligned}$$

We have

$$\left(\sum_{i=0}^k A^i \right) b = (I - A)^{-1}b$$

Linear Algorithms

Write

$$x_0 = \sum_{j=1}^d a_j q_j$$

where $(q_j)_{j=1}^d$ is the eigenbasis of $\mathbb{R}^d$. Then

$$\lim_{k \rightarrow \infty} A^k x_0 = A^k \sum_{j=1}^d q_j = \sum_{j=1}^d a_j A^k q_j = \sum_{j=1}^d a_j \lambda_j^k q_j$$

Linear Algorithms

Design algorithm for solving a linear system $Ax = b$ where A is positive definite?

Stochastic Algorithms

What do do if we can only measure

$$\hat{T}(x) = T(x) + n$$

where n is some noise?

Stochastic Algorithms

Can we do

$$x_{k+1} = \hat{T}(x_k)$$

Stochastic Algorithms

Can we do

$$x_{k+1} = \hat{T}(x_k)$$

Suppose that α -contraction and that n_k are independent with $\mathbb{E}||n_k||^2 = \sigma^2$

$$\begin{aligned} \mathbb{E}||x_{k+1} - x^*|| &\leq ||T(x_k) - x^*|| + \mathbb{E}||n_k||^2 \\ &\leq \alpha ||x_k - x^*|| + \sigma^2 \\ &\leq \alpha^{k+1} ||x_k - x^*|| + \sigma^2 \sum_{i=0}^k \alpha^i \\ &\leq \underbrace{\alpha^{k+1} ||x_k - x^*||}_{\rightarrow 0} + \underbrace{\frac{\sigma^2}{1 - \alpha}}_{\text{constant}} \end{aligned}$$

Stochastic Algorithms

Can we do better?

Stochastic Algorithms

Can we do better?

Idea

Use soft update with step-size $\gamma \in [0, 1]$

$$x_{k+1} = x_k + \gamma(\hat{T}(x_k) - x_k)$$

Stochastic Algorithms

Can we do better?

Idea

Use soft update with step-size $\gamma \in [0, 1]$

$$x_{k+1} = x_k + \gamma(\hat{T}(x_k) - x_k)$$

can also be written as convex combination of x_k and $\hat{T}(x_k)$

$$x_{k+1} = (1 - \gamma)x_k + \gamma\hat{T}(x_k)$$

Stochastic Algorithms

Question

If $T(x)$ is α -contraction with fixed point x^* , what can we tell about the fixed points of:

$$x_{k+1} = (1 - \gamma)x_k + \gamma T(x_k)$$

Stochastic Algorithms

Question

If $T(x)$ is α -contraction with fixed point x^* , what can we tell about the fixed points of:

$$x_{k+1} = (1 - \gamma)x_k + \gamma T(x_k)$$

If $T(x^*) = x^*$ then we have

$$(1 - \gamma)x^* + \gamma T(x^*) = (1 - \gamma)x^* + \gamma x^* = x^*$$

Stochastic Algorithms

Question

If $T(x)$ is α -contraction with fixed point x^* is the algorithm

$$x_{k+1} = (1 - \gamma)x_k + \gamma T(x_k)$$

also an α -contraction?

Stochastic Algorithms

Question

If $T(x)$ is α -contraction with fixed point x^* is the algorithm

$$x_{k+1} = (1 - \gamma)x_k + \gamma T(x_k)$$

also an α -contraction?

We have

$$\begin{aligned} \|(1 - \gamma)x + \gamma T(x) - x^*\| &\leq (1 - \gamma)\|(1 - \gamma)x - x^*\| + \gamma\|T(x) - x^*\| \\ &\leq (1 - \gamma)\|x - x^*\| + \gamma\alpha\|x - x^*\| \\ &\leq [(1 - \gamma) + \gamma\alpha]\|x - x^*\| \end{aligned}$$

Stochastic Algorithms

Question

What happens if we have noise

$$x_{k+1} = (1 - \gamma)x_k + \gamma \hat{T}(x_k)$$

Stochastic Algorithms

Question

What happens if we have noise

$$x_{k+1} = (1 - \gamma)x_k + \gamma \hat{T}(x_k)$$

See homework problem 7.